STUDENT ACTIVITY

Python Fundamentals for Data Science — Activity

Activity: Code Analysis and Output Prediction

Q: Read the three Python code snippets below. For each snippet, perform the requested analysis based on your knowledge of Python syntax and operations.

Next, write three to four sentences explaining why Python is preferred for Data Science, and name two libraries specifically used for data analysis and visualization.

The Snippets:

1. `age = 21` **and** `student = {"name": "Sahil"}`**.** Identify the specific built-in data types for both of these variables.
2. Review the following logic: `marks = 85` `if marks >= 90:` `print("Excellent")` `elif marks >= 75:` `print("Good")` Predict the exact output that will be printed.
3. Review the following logic: `count = 1` `while count <= 5:` `print(count)` `count += 1` Identify the type of loop used and state the final number that will be printed before the loop terminates.

1. Analyze the Python Code Snippets

Snippet 1

age = 21

student = {"name": "Sahil"}

Identify the built-in data types

Variable	Value	Built-in Data Type
age	21	**int (Integer)**
student	{"name": "Sahil"}	**dict (Dictionary)**

Explanation

- **age** stores a whole number (`21`), so its data type is **int**.
- **student** stores data as a key-value pair (`"name": "Sahil"`), so its data type is **dictionary (dict)**.

Snippet 2

```
marks = 85

if marks >= 90:

   print("Excellent")

elif marks >= 75:

   print("Good")
```

Explanation

- The value of marks is **85.**
- The first condition (marks >= 90) is **False**.
- The second condition (marks >= 75) is **True**.
- Therefore, the program prints:

Snippet 3

```
count = 1

while count <= 5:

   print(count)

   count += 1
```

Loop Type - While Loop
Output

1
2
3
4
5

Explanation

- The while loop executes as long as count <= 5.
- The variable count starts at **1** and increases by **1** in each iteration.
- The last value printed before the loop terminates is **5**.

2. Why is Python Preferred for Data Science?

Python is widely preferred for Data Science because it is easy to learn, has simple syntax, and offers a large ecosystem of powerful libraries. It enables efficient data collection, cleaning, analysis, visualization, and machine learning, making it suitable for handling complex datasets. Python also has strong community support and integrates well with databases, cloud platforms, and big data tools. Its versatility and extensive libraries make it one of the most popular programming languages for data science applications.

3. Two Python Libraries Used for Data Analysis and Visualization

Purpose	Library	Description
Data Analysis	**Pandas**	Used for data manipulation, cleaning, filtering, grouping, and analyzing structured datasets using DataFrames.
Data Visualization	**Matplotlib**	Used to create charts and graphs such as line charts, bar charts, histograms, scatter plots, and pie charts for visual data analysis.

Final Answers Summary

Question	Answer
Snippet 1	`age` → **int**, `student` → **dict**
Snippet 2 Output	**Good**
Snippet 3 Loop Type	**While Loop**
Final Number Printed	**5**
Why Python for Data Science?	Easy syntax, powerful libraries, efficient data analysis, strong community support, and excellent machine learning capabilities.
Data Analysis Library	**Pandas**
Visualization Library	**Matplotlib**